



Quantum Optimization for the Vehicle Routing Problem

IBM

Qiskit

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Vehicle Routing Problem (VRP) is a concept in the field of transportation that involves identifying the most efficient route for a set of customers with fixed positions [TSP derived problem]

→ **NP-hard problem** with a non-polynomial time complexity solution

(Khan, Alghamdi, & Khan, 2023)

QAOA is a quantum (hybrid quantum-classical) algorithm that aims to approximate solutions to combinatorial optimization problems.

(Farhi et al., 2014)

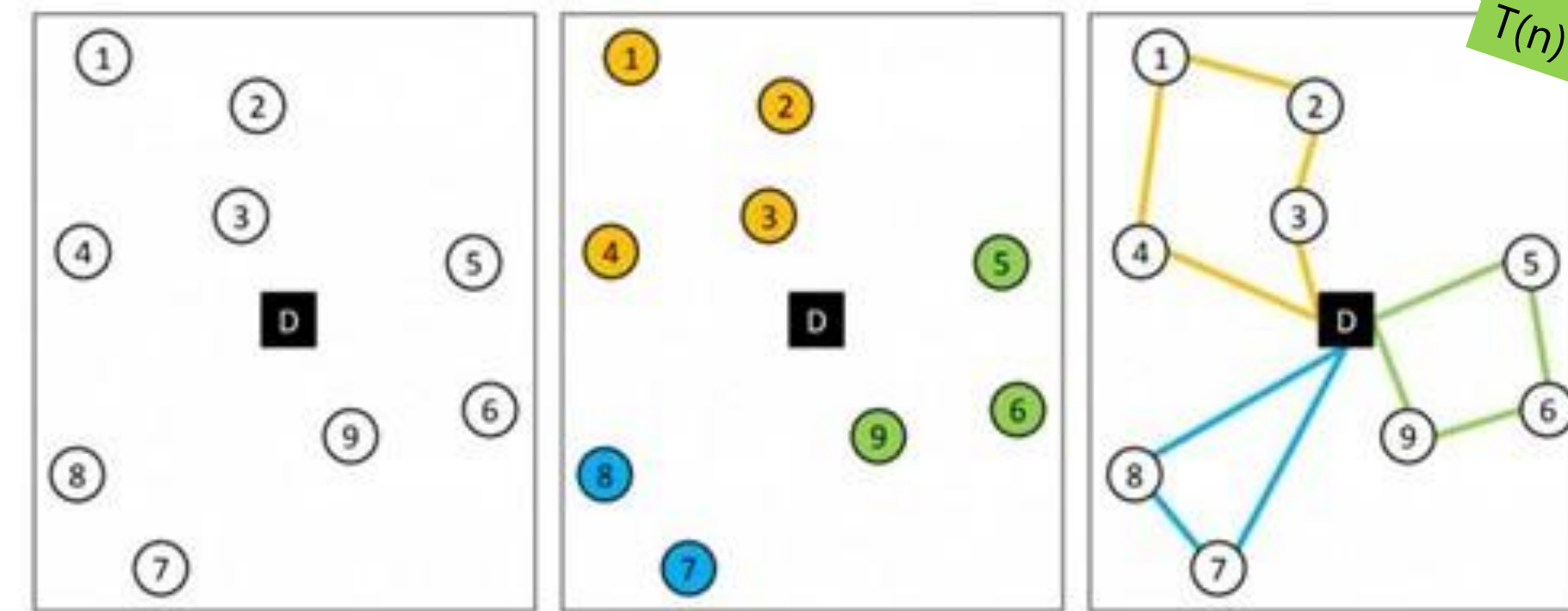
- ✓ **Qubits** (represent the problem and output the solution)
- ✓ **Hamiltonian and Cost Function** with trainable parameters (as an objective function)
- ✓ **Quantum Circuits** (that we will use to iterate through the parameters)

Why does quantum need so many qubits?

What happens when circuits get deeper?

Research Questions

- RQ1 How can the spatial data of VRP be converted and formulated into a structure that is solvable by quantum algorithms, by efficiently allocate qubits?
- RQ2 To what extent are quantum algorithms effective compared to best-performing classical VRP solvers?



NP-Hard!
 $T(n) = \Omega(n!)$

Applications

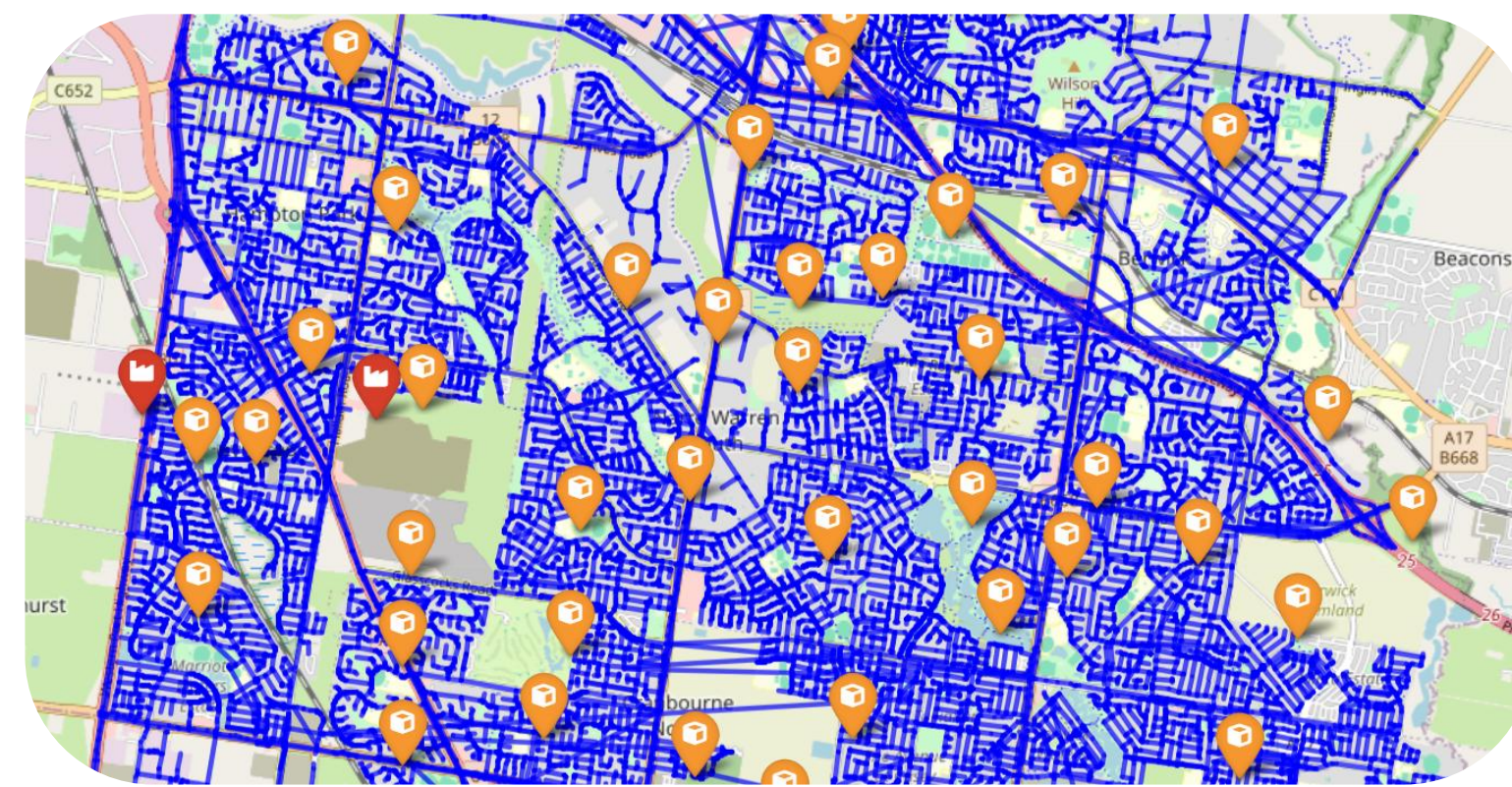
- Logistics & Delivery
- Waste Collection
- School Bus Routing

Can it beat classical optimization?

1

Case Study: Waste Collection in City of Casey

- Depots: Waste collection operations points (data.casey.vic.gov.au)
- Customers: Public litter bins' location (data.casey.vic.gov.au)
- Routes: Victoria's road network (vicdata.vicroads.vic.gov.au)
- Constraints:**
 - Traffic level (opendata.arcgis.com)
 - Weather conditions (data.casey.vic.gov.au)
 - Road Crashes (discover.data.vic.gov.au)



2

Encoding: From Classical to Quantum

Encoding Method	Qubit Requirement	Number of Qubits	Cost Function Simplicity
Naïve Adjacency (Azad et al., 2020 Yasumura et al., 2025)	N^2	Large	Simple
Logarithmic Permutation Theoretical minimum	$\lceil \log_2(n!) \rceil$	Small	Complex (impractical)
Node-Visit-Time (NEW) This Research's Contribution	$n \cdot \lceil \log_2(N) \rceil$	Moderate	Fair

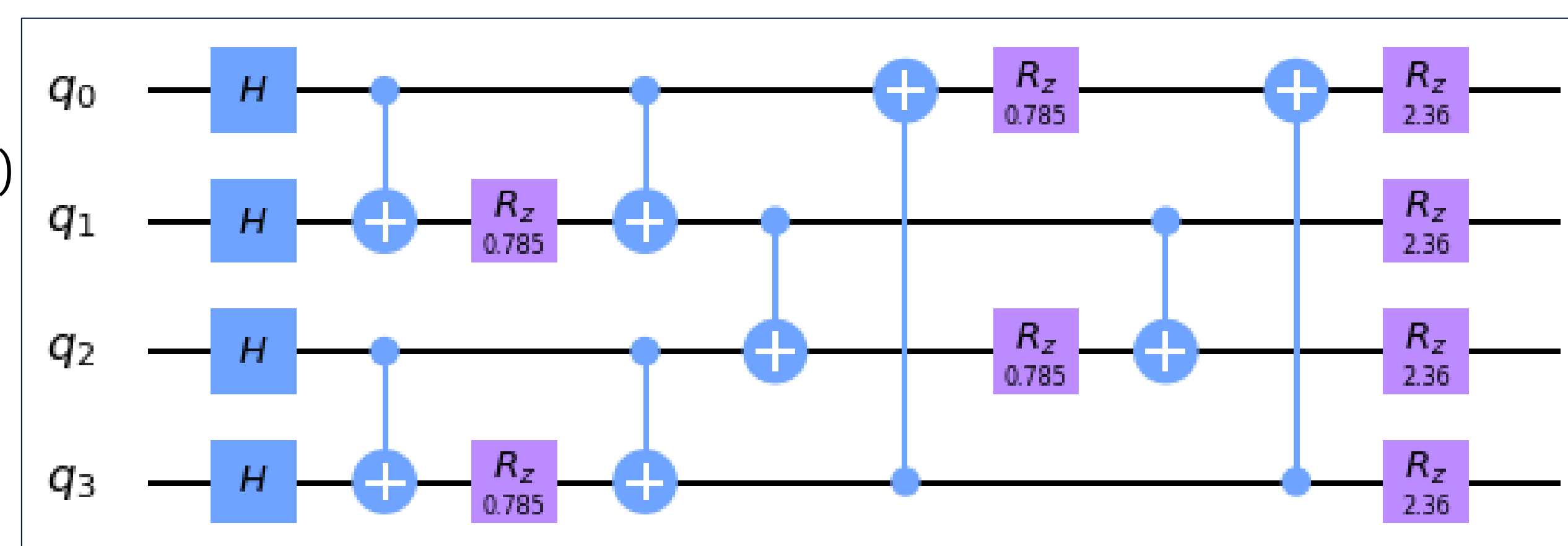
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QAOA: Quantum Approximate Optimization Algorithm

Optimizers: COBYLA (a gradient-free optimizer)

- Gradient-Free Optimizer**
 - **Justification:** QAOA parameter landscape is highly non-convex and noisy due to (NISQ era).
 - **Robustness:** Avoids getting stuck in plateaus (where there are no gradients)
- COBYLA's advantage and drawback**
 - **Advantage:** designed to handle non-linear optimization (matched with VRP)
 - **Drawback:** slow to converge

Quantum Circuits
(for single depth QAOA)



4

Optimality Ratio

the best solution (most optimal/minimal path) divided by the expectation value of QAOA circuit (the path distance output)

$$\text{Optimality Ratio} = \frac{C_{\min}}{\langle C \rangle}$$

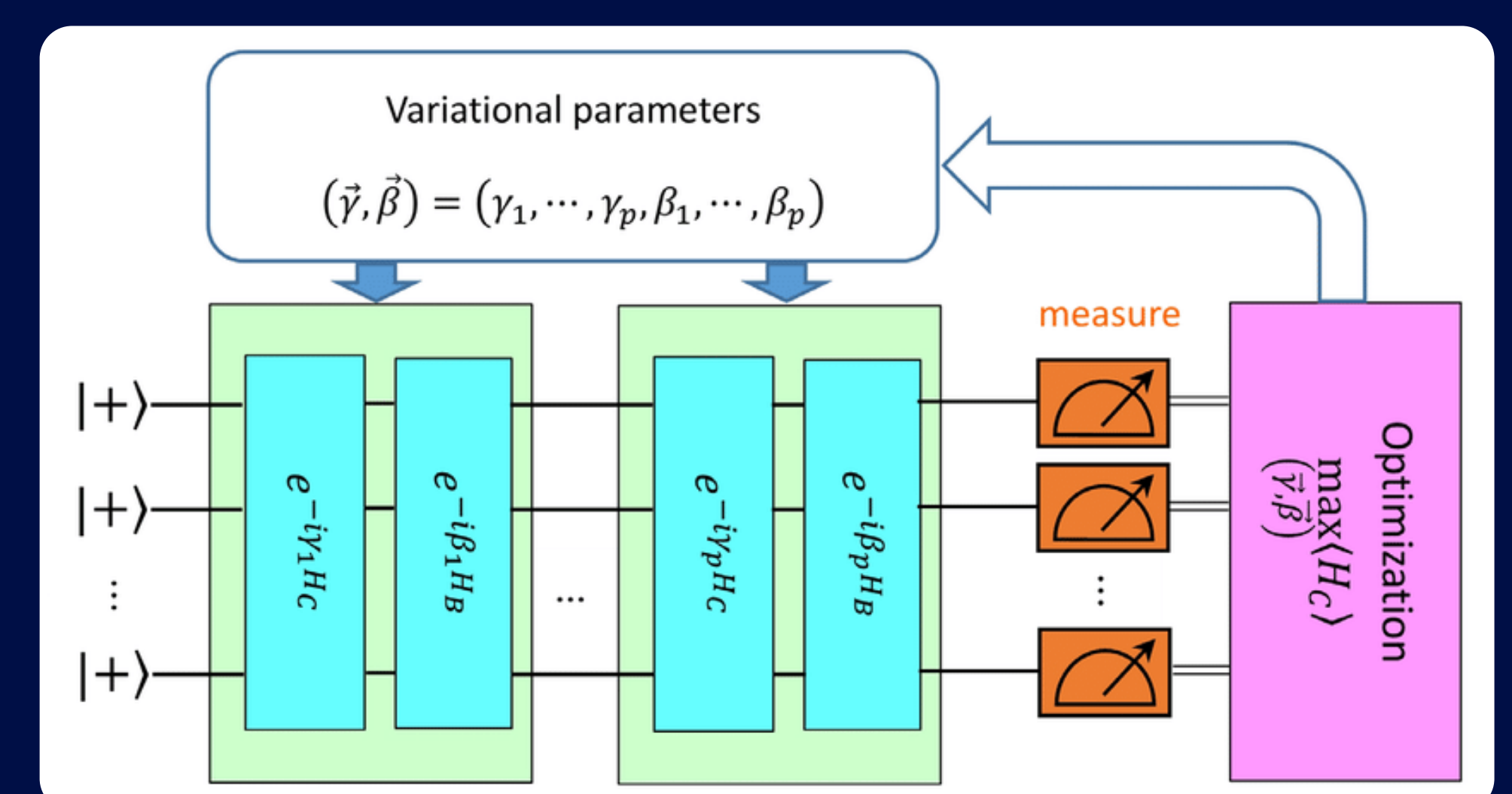
technique	avg_opt_ratio	std_opt_ratio
naive_adjacency	0.987150	0.027566
node_visit_time	0.928273	0.078774

Methodology and Result

- Naïve Encoding** (large number of qubits)
transform each cell in naïve adjacency matrix data into a binary qubit representation
For N nodes, there will be N^2 distances, and each will be mapped to a qubit using a one-hot encoding manner
- Logarithmic Encoding** (theoretical minimum)
each VRP path (total of around N! combinations of paths) is encoded into a binary system of encoding
impractical: doing the encoding has the same work as doing brute-force solving
- Node-Visit-Time Encoding (NEW)**
each node will be represented by the binary (such as 00, 01, 10, 11) and concatenated side-by-side
For instance, with `0 → 2 → 1 → 3` will be mapped to `00100111`

QAOA Algorithm's Schema

a hybrid of quantum and classical algorithm using optimizers which iterates through the quantum circuits



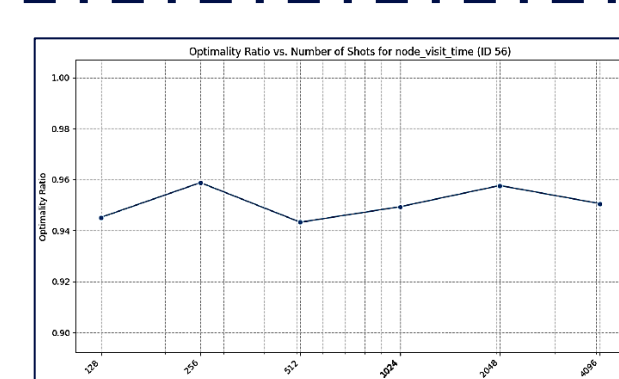
Contribution

- Qubit efficient way to solve the VRP
Trade-off between number of qubit and time complexity
As quantum computing hardware is costly
- Reduced Time Complexity
 $T(n, d, v)_{\text{Classical}} \approx (n + d)! \cdot v^n$
 $T(n, d, v, p, s)_{\text{Quantum}} \approx s \cdot p \cdot v \cdot (n + d)^2$

What Lies Ahead

- Can quantum computers solve city-wide delivery problems faster than classical?
- What if a few qubits could optimize hundreds of routes at once?
- Will QAOA become the next standard for logistics optimization?
- How far can quantum algorithms push real-world delivery efficiency?

"currently classical still beats quantum"



Technique	Success Attempts	Optimal Attempts	Failed Attempts
Naive Adjacency	10	8	13
Node Visit Time	21	4	2

